

Low-Complexity Detection of Multiple Moving Targets with Distributed Radar System

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Abstract—Addressing the high computational and communication overheads for multi-target detection in distributed radar systems, this paper proposes a low-complexity two-step method for efficiently detecting multiple moving targets. First, local nodes apply a low-threshold detection to identify and then transmit the potential target range-Doppler (R-D) cells, thereby significantly reducing the communication burden. Then, the fusion center constructs a spatial grid map and employs a multi-mapping vector cascade voting detection mechanism, which can avoid exhaustive parameter search, and further reduce the computational cost. Finally, closed-form expressions for system performance metrics are derived. Simulations demonstrate that, compared to the traditional full-data fusion benchmark, the proposed method reduces the data size required for transmission to about 5% and the computational cost by a factor of 10^5 , while resulting in an approximate 1 dB signal-to-noise ratio (SNR) loss. Moreover, the proposed multi-mapping vector cascade voting mechanism effectively suppresses ghost targets.

Index Terms—Distributed radar, data fusion, multi-moving target detection, low-complexity detection.

I. INTRODUCTION

Distributed radar architectures have provided new avenues for modern radar systems [1], [2], offering enhanced sensing via cooperative observation. Compared to monostatic radar, its multi-perspective nature improves both positioning accuracy and detection performance [3], [4], while spatial diversity suppresses Radar cross-sectional scintillation for greater stability [5]–[9]. Such system also exhibits inherent anti-jamming advantages in complex electromagnetic environments [10]. While conventional moving target detection relies on Doppler processing at a single node [11]–[13], effectively applying these techniques to a distributed system with multiple perspectives faces significant difficulty. There are two core challenges: i) the substantial communication burden from transmitting full

R-D data to the fusion center [14]; ii) the spatio-temporal misalignment issues arising from geographically dispersed nodes, which complicate data fusion [15], [16].

To address the communication bottleneck, existing works have employed local threshold censoring [17], [18] or quantization techniques [19], [20] to reduce data size. Some innovative approaches transmit only the maximum Doppler value per range cell as “best-matched data” [21], [22]. To tackle data heterogeneity, spatial grid mapping methods project signals onto discrete grids to unify coordinates [23]–[27]. However, these methods have limitations: the maximum-value strategy may discard weak target information, while full-data transmission requires exhaustive search with excessive computational cost. Moreover, under high SNR conditions, the full-data fusion approach is prone to generating ghost targets.

To overcome the aforementioned limitations, this paper proposes a two-stage detection method, which achieves partial suppression of ghost targets and joint optimization of communication and computational overheads simultaneously. Specifically, local nodes perform initial detection with a low threshold to preserve potential targets, and only transmit the detected R-D cells to reduce data traffic. At the fusion center, target range and Doppler information from all nodes are mapped onto a spatial grid, thereby aligning the measurements across nodes within a unified coordinate system. A cascade voting mechanism then validates candidate target grid cells and velocity-angle combinations by requiring consistent detections across multiple channels, which avoids false detections caused by sporadic high-energy returns in individual channels and effectively suppressing ghost targets under high SNR conditions. Energy accumulation and maximum-value search are performed only on high-confidence candidates and the maximum-energy state is taken as the final estimate. Theoretically, closed-form expressions for communication and computational overheads, detection probability, and false alarm probability are derived, based on which we determine the optimal voting

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threshold and local false alarm probability.

II. SIGNAL MODEL

Consider a distributed radar system with N_t transmitters and N_r receivers. Each transmitter emits an orthogonal waveform $\Phi_m(t)$ over K pulses in a Coherent Processing Interval, where the waveforms satisfy $\int_0^{T_p} \Phi_{m_1}(t) \Phi_{m_2}^*(t-\tau) dt \approx 0$ for $m_1 \neq m_2$. The signal transmitted by the m -th ($m = 1, 2, \dots, N_t$) transmitter during the k -th ($k = 1, 2, \dots, K$) pulse is given by

$$s_{m,k}(t) = \text{rect}\left(\frac{t}{T_p}\right) \Phi_m(t) e^{j2\pi f_c(t+(k-1)T_r)}, \quad (1)$$

where T_r denotes the pulse repetition interval, T_p is the pulse width, f_c is the carrier frequency, and $\text{rect}(\cdot)$ represents the rectangular window function.

After applying matched filtering to the echo and sampling at a rate of T_s , the echo received by the n -th ($n = 1, 2, \dots, N_r$) receiver from the m -th transmitter for the k -th pulse and l -th ($l = 1, 2, \dots, L$) sample is

$$x_{m,n}^k(l) = \alpha_{m,n}(l) e^{-j2\pi f_d^{m,n}(k-1)T_r} + w_{m,n}^k(l), \quad (2)$$

where $f_d^{m,n}$ is the Doppler shift, $w_{m,n}^k(l) \sim \mathcal{CN}(0, \sigma_w^2)$ is complex additive white Gaussian noise (AWGN), and the term $\alpha_{m,n}(l)$ is the complex backscattering coefficient at the l -th range cell, encompasses the target radar cross section, the bistatic path loss, and the matched filter gain.

For a single range cell, the echoes from all K pulses form an observation vector $\mathbf{x}_{\tilde{k}}$ for the $\tilde{k}$ -th Doppler channel. Target detection can then be formulated as a binary hypothesis test

$$\begin{cases} \mathcal{H}_0 : \mathbf{x}_{\tilde{k}} = \mathbf{w}_{\tilde{k}} \\ \mathcal{H}_1 : \mathbf{x}_{\tilde{k}} = \xi_{\tilde{k}} \mathbf{d}_{\tilde{k}} + \mathbf{w}_{\tilde{k}}, \end{cases} \quad (3)$$

where $\xi_{\tilde{k}} = \alpha_{m,n}(\tilde{k})$, $\mathbf{d}_{\tilde{k}} = [1, \dots, e^{j2\pi f_d^{m,n}(K-1)T_r}]^\top$ is the Doppler vector, $\mathbf{w}_{\tilde{k}} = [w_{m,n}^1(\tilde{k}), \dots, w_{m,n}^K(\tilde{k})]^\top$.

III. DATA FUSION AND TWO-STEP DETECTOR DESIGN

The core challenge in distributed radar is to detect targets' spatial and kinematic states under stringent communication and computational constraints, as heterogeneous data from spatially separated nodes limit the effectiveness of conventional detection methods. To address this challenge, we propose a unified grid-based fusion framework that systematically handles data heterogeneity in both range and Doppler dimensions. This framework is combined with a two-step detection strategy, incorporating local threshold detection and multi-mapping vector cascade voting detection, which may significantly reduce both communication and computational overheads.

A. Grid-based Range-Doppler Fusion

The surveillance area is divided into discrete grid cells G , enabling consistent data association across all channels. The principle of range fusion is to project each grid cell onto the range axis of every channel. A target residing in the same grid cell $G(x_0, y_0)$ will appear in different range bins $R_{m,n}$

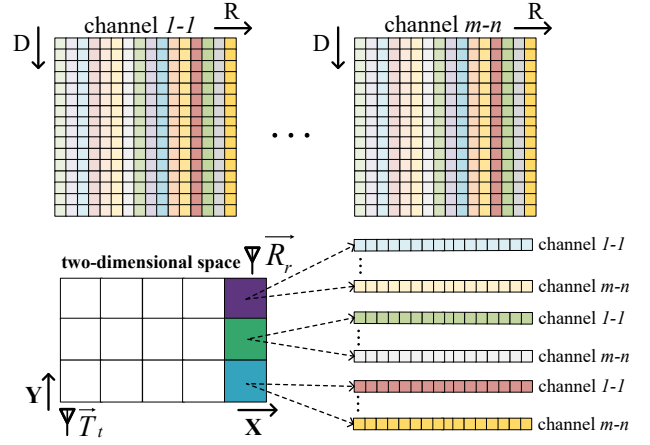


Fig. 1: Mapping from Grid Space to Range Units

across different transmitter-receiver pairs (m, n) , as illustrated in Fig.1. For a transmitter at $T_t(x_m^t, y_m^t)$ and a receiver at $T_r(x_n^r, y_n^r)$, the n_g -th spatial grid maps onto the distance unit $R_{m,n}$ of channel (m, n)

$$R_{m,n} = \frac{1}{2} \left(\sqrt{(x_0 - x_m^t)^2 + (y_0 - y_m^t)^2} + \sqrt{(x_0 - x_n^r)^2 + (y_0 - y_n^r)^2} \right). \quad (4)$$

The obtained spatial distance mapping matrix $\mathbf{z}_{m,n}^{R,n_g} \in \mathbb{R}^{1 \times K}$ is described as

$$\mathbf{z}_{m,n}^{R,n_g} = \mathbf{F}_{m,n}^{R,n_g} \mathbf{E}_{m,n}, \quad (5)$$

where $\mathbf{F}_{m,n}^{R,n_g} = [0, \dots, 1, \dots, 0] \in \mathbb{R}^{1 \times L}$ is the range-cell selection matrix, and $\mathbf{E}_{m,n} \in \mathbb{R}^{L \times K}$ denotes the R-D energy matrix of the channel (m, n) . By stacking the distance mapping matrices from all channels into a complete vector, i.e., $\mathbf{Z}^{R,n_g} \in \mathbb{R}^{N_t N_r \times K}$, we have that

$$\mathbf{Z}^{R,n_g} = \begin{bmatrix} \mathbf{z}_{1,1}^{R,n_g} & \mathbf{z}_{1,2}^{R,n_g} & \dots & \mathbf{z}_{N_t, N_r}^{R,n_g} \end{bmatrix}^T. \quad (6)$$

For Doppler fusion, we estimate the target's velocity vector by testing hypotheses over a discretized velocity-angle (v, θ) state, as shown in Fig.2. For each grid cell, we enumerate candidate (v, θ) pairs. The theoretical Doppler frequency for channel (m, n) is given by

$$f_{m,n} = \frac{f_c v}{c} [\cos(\theta_m^t - \theta) + \cos(\theta_n^r - \theta)], \quad (7)$$

where θ_m^t and θ_n^r are the angles of the transmitter and receiver relative to the grid cell center. The motion state mapping matrix $\mathbf{z}_{m,n}^{D,n_g,n_s} \in \mathbb{R}$ for the n_s -th motion state of the n_g -th grid cell in channel (m, n) is obtained as

$$\mathbf{z}_{m,n}^{D,n_g,n_s} = \mathbf{F}_{m,n}^{D,n_g,n_s} (\mathbf{z}_{m,n}^{R,n_g})^T, \quad (8)$$

where $\mathbf{F}_{m,n}^{D,n_g,n_s} = [0, \dots, 1, \dots, 0] \in \mathbb{R}^{1 \times K}$ is the Doppler state selection matrix. Stacking the motion state mapping matrices from all channels into a complete vector, i.e., $\mathbf{Z}^{D,n_g,n_s} \in \mathbb{R}^{N_t N_r \times 1}$, yields

$$\mathbf{Z}^{D,n_g,n_s} = \begin{bmatrix} \mathbf{z}_{1,1}^{D,n_g,n_s} & \mathbf{z}_{1,2}^{D,n_g,n_s} & \dots & \mathbf{z}_{N_t, N_r}^{D,n_g,n_s} \end{bmatrix}^T. \quad (9)$$

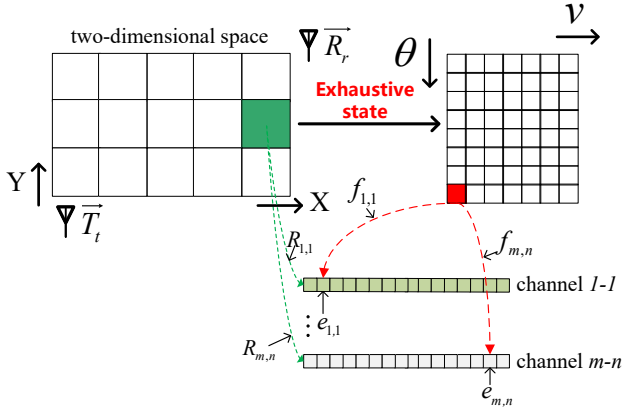


Fig. 2: Mapping of (v, θ) State to Doppler Dimension

Subsequently, the state with the maximum energy is searched among the N_s candidate states $\left[\left\| \mathbf{Z}^{D,n_g,1} \right\|^2, \dots, \left\| \mathbf{Z}^{D,n_g,N_s} \right\|^2 \right]$, as follows

$$(\hat{v}_0, \hat{\theta}_0) = \arg \max_{n_s} \left[\left\| \mathbf{Z}^{D,n_g,1} \right\|^2, \dots, \left\| \mathbf{Z}^{D,n_g,N_s} \right\|^2 \right], \quad (10)$$

where, $(\hat{v}_0, \hat{\theta}_0)$ represents the estimated target state in the n_g -th grid cell, with an energy of $\left\| \mathbf{Z}^{D,n_g,n_s} \right\|^2$.

B. Two-step Detector Design for Low-Complexity Fusion

The full data fusion strategy is computationally prohibitive. To reduce the cost, we propose a two-stage detection scheme.

Step 1: Local Data Censoring. Each channel employs a Generalized Likelihood Ratio Test (GLRT) detector to filter out noise-dominated cells. The test statistic is

$$T(\mathbf{x}_{\tilde{\mathbf{k}}}) = \frac{|\mathbf{d}_{\tilde{\mathbf{k}}}^H \mathbf{x}_{\tilde{\mathbf{k}}}|^2}{\mathbf{d}_{\tilde{\mathbf{k}}}^H \mathbf{d}_{\tilde{\mathbf{k}}}} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\geq}} \Lambda. \quad (11)$$

The local detector transmits to the fusion center only the statistics of those cells whose values exceed the threshold Λ , along with their positional indices in the R-D map.

Step 2: Fusion Center Decision. The fusion center implements a multi-mapping vector cascade voting detection rule to avoid processing unlikely target candidates. The cascade voting consists of a spatial grid cell validity vote

$$G_{\text{valid}}^{n_g} = \begin{cases} 1, & \text{rank}(\mathbf{Z}^{R,n_g}) \geq \kappa \\ 0, & \text{otherwise} \end{cases} \quad (12)$$

and a target motion state validity vote

$$S_{\text{valid}}^{n_s} = \begin{cases} 1, & \left\| \mathbf{Z}^{D,n_g,n_s} \right\|_0 \geq \kappa \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

where $\text{rank}(\cdot)$ denotes the matrix rank, $\|\cdot\|_0$ the L0-norm, and κ a positive integer for the minimum required multi-channel votes. The motion state validity vote is triggered only if the current spatial grid cell passes the validity vote; fusion and search are performed solely for valid motion states in valid

spatial grid cells to greatly cut computational cost. The target existence is determined by the criterion: $G_{\text{valid}}^{n_g} \cdot S_{\text{valid}}^{n_s} = 1$.

IV. SYSTEM PERFORMANCE ANALYSIS

The fusion center receiving data from local nodes and executing the multi-mapping vector cascade voting for final decision-making. This section provides a comprehensive analysis of the resulting system performance. We calculated the communication burden from the local nodes and the computational cost at the fusion center. Furthermore, we characterize the detection and false alarm probabilities.

A. Communication and Computation Burden

Under the independent and identically distributed assumption for all observation channels, we define the size of the per-channel R-D matrix as $L \times K$ (each data point occupies B bytes). Let N_g denote the number of grid cells, N_s the number of (v, θ) pairs. Without optimization, the raw per-channel data size is $Q_c = L \times K \times B$. After local GLRT detection with threshold Λ and local false alarm probability P_{fa}^{local} , the expected per-channel transmitted data size reduces to

$$\mathbb{E}[Q_c^{\text{trans}}] = \lceil Q_c \cdot P_{fa}^{\text{local}} \rceil. \quad (14)$$

At the fusion center, the multi-mapping vector cascade voting detection further reduces the number of grid cells and parameter combinations that require processing. The probability that $G_{\text{valid}}^{n_g} = 1$ is given by

$$P_g^{\text{val}} = \sum_{i=\kappa}^N \binom{N}{i} P_0^i (1 - P_0)^{N-i}, \quad (15)$$

where $N = N_t N_r$ and $P_0 = 1 - (1 - P_{fa}^{\text{local}})^K$.

Similarly, the probability that $S_{\text{valid}}^{n_s} = 1$ is given by

$$P_s^{\text{val}} = \sum_{j=\kappa}^N \binom{N}{j} (P_{fa}^{\text{local}})^j (1 - P_{fa}^{\text{local}})^{N-j}. \quad (16)$$

The proposed method has a computational complexity of $\mathcal{O}(N_g P_g^{\text{val}} N_s P_s^{\text{val}})$, a reduction from the $\mathcal{O}(N_g N_s)$ full-search baseline.

B. System Detection and False Alarm Probability

The local detection probability P_d^{local} for a single channel under AWGN is given by the Marcum Q-function, specifically

$$P_d^{\text{local}} = Q\left(\sqrt{2 \cdot \text{SNR}}, \sqrt{-2 \ln(P_{fa}^{\text{local}})}\right). \quad (17)$$

The system-level detection probability P_d^{sys} is approximated by the probability that the true target parameters (v_0, θ_0) are voted for by at least κ channels, that is

$$P_d^{\text{sys}} = \sum_{j=\kappa}^N \binom{N}{j} (P_d^{\text{local}})^j (1 - P_d^{\text{local}})^{N-j}. \quad (18)$$

The system-level false alarm probability P_{fa}^{sys} considers the probability that at least one incorrect (v, θ) combination is validated in at least one grid cell, that is,

$$P_{fa}^{\text{sys}} = 1 - (1 - P_s^{\text{val}})^{N_s}. \quad (19)$$

V. SIMULATION RESULTS

Numerical simulations validate the proposed method using a distributed radar system with $N_t = 4$ transmitters and $N_r = 4$ receivers ($N = 16$ channels). Key parameters include: $N_g = 2916$ grid cells, $N_s = 2304$ (v, θ) combinations, $K = 128$ pulses, and $L = 1024$ samples per pulse.

To evaluate the trade-off between detection performance and system overhead, we analyze the impact of the voting threshold κ under a fixed global false alarm probability $P_{fa}^{sys} = 10^{-6}$. Based on the relationships defined in (14) and (19), each κ corresponds to a specific local false alarm probability P_{fa}^{local} . As jointly shown in Fig. 3a and Fig. 3b, increasing κ raises both the communication and computational overheads at the fusion center. The detection probability governed by (17) and (18) is then evaluated for the P_{fa}^{local} values corresponding to different κ (given in Fig. 3a). The results indicate that the system achieves its peak detection performance when $\kappa = 10$, thereby balancing performance with resource constraints.

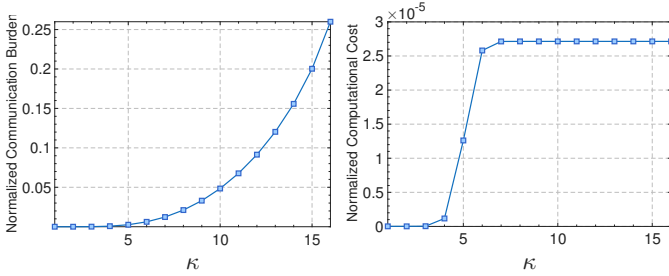


Fig. 3: Communication and computation analysis

Fig. 4 presents the Monte Carlo simulation results comparing the detection performance of three methods: i) the proposed method, ii) the full-data fusion method, and iii) the local threshold method. Results show that the proposed method incurs about 1 dB SNR loss compared to the full-data fusion method, while outperforming the local threshold method. This loss stems from the fact that the multi-mapping vector cascade voting makes decisions solely based on single-channel results, where no incoherent gain is generated from the energy across different channels; and it is precisely this characteristic that endows the proposed method with ghost target suppression performance. Fig.5 compares the normalized communication and computational costs. The communication overhead of the proposed method is similar to that of the local threshold method and lower than that of full-data fusion, while its computational cost is significantly lower than both alternatives.

Finally, Fig.6 visually compares the data fusion results. Under high SNR conditions, the full-data fusion method often exhibits numerous ghost targets, whereas the proposed multi-mapping vector cascade voting method effectively mitigates this problem.

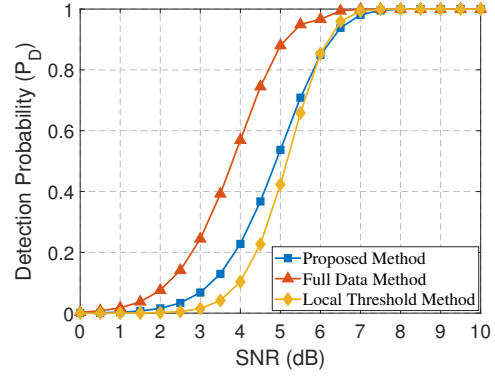


Fig. 4: Detection probability comparison of three schemes

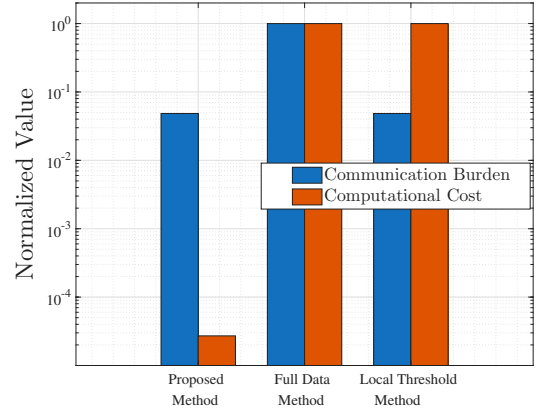


Fig. 5: Comparison of communication and computing costs

VI. CONCLUSION

This paper proposes a low-complexity multi-target detection framework for resource-constrained distributed radars, combining R-D grid mapping with a two-step detection scheme: local low-threshold primary detection for data compression, and multi-mapping cascade voting at the fusion center for candidate censoring. At a $P_{fa}^{sys} = 10^{-6}$, numerical results show it reduces data transmission to 5% of the original, cuts computational cost by 10^5 , with 1 dB SNR degradation versus full-data fusion, while effectively suppressing ghost targets via the cascade voting design.

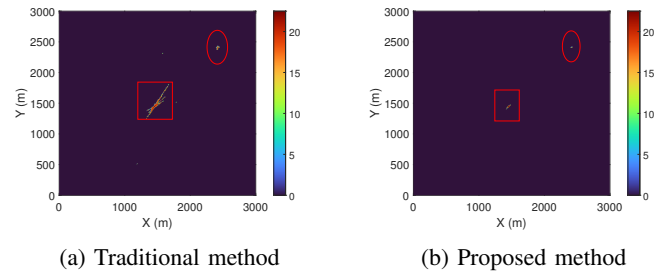


Fig. 6: Comparison of data fusion results($SNR = 15$ dB)

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